

Product Requirements Document

Fabric Migration Confidence Center

1. Overview

Fabric Migration Confidence Center is a product concept for enterprise customers migrating from Azure Synapse Analytics to Microsoft Fabric.

The product does not replace Microsoft's existing migration tools. Instead, it creates a unified readiness, blocker, remediation, and validation layer around them.

The core idea is simple:

Customers do not delay migration only because they lack a migration button. They delay because they cannot confidently answer what will break, how much work is left, and whether the migration worked after cutover.

2. Problem statement

Microsoft Fabric already provides migration tooling for specific workloads. There is a Fabric Data Warehouse Migration Assistant, a Synapse Spark to Fabric Spark Migration Assistant, and a Synapse pipeline migration experience.

But enterprise migration is not limited to one workload.

A large Synapse customer may have:

- Dedicated SQL pools
- Serverless SQL views
- Spark pools
- Notebooks
- Spark job definitions
- Pipelines
- Linked services
- External tables
- Key Vault references
- Private networking rules
- CI/CD workflows
- Power BI reports
- Monitoring and alerting processes

These assets depend on each other. A pipeline may call a notebook. That notebook may depend on storage access. A report may depend on an external table. A linked service may depend on a secret. A private endpoint rule may affect whether the migrated workload can run in production.

Today, customers must understand these risks across several tools, docs, exports, scripts, and manual reviews. This creates uncertainty before cutover.

3. Product thesis

Microsoft has workload-specific migration tools. The missing opportunity is a cross-workload confidence layer.

Fabric Migration Confidence Center helps enterprise migration teams answer:

- What exists in our Synapse estate?
- What depends on what?
- Which assets are ready to migrate?
- Which assets need review?
- Which assets are blocked?
- Which blockers are Microsoft roadmap gaps, preview limitations, or customer-side remediation work?
- What should we fix first?
- What proof do we have that migration worked?

4. Target users

Primary users

Enterprise Data Platform Lead

Owns the company's data platform strategy and needs to know if Fabric migration is safe.

Data Engineering Manager

Owns pipelines, notebooks, SQL objects, Spark workloads, and migration execution.

Cloud Migration Program Manager

Owns timelines, blockers, readiness reporting, stakeholder communication, and cutover planning.

Azure/Fabric Solution Architect

Needs to guide customers through technical readiness and migration design.

Microsoft Customer Success or FastTrack Engineer

Needs a clear way to identify blocked customers and guide them toward Fabric adoption.

Security and Governance Lead

Needs confidence that private networking, Key Vault, identity, and access patterns still work.

BI or Analytics Lead

Needs to know whether Power BI reports and semantic models will continue to refresh after migration.

5. Jobs to be done

Job 1: Discover the estate

When I begin a Synapse-to-Fabric migration, I want to see every relevant asset across my estate so that I do not miss hidden dependencies.

Job 2: Understand dependency risk

When I review migration readiness, I want to see which assets depend on each other so that I can avoid breaking downstream pipelines, reports, or security flows.

Job 3: Prioritize blockers

When blockers are found, I want them grouped by risk and action type so that my team knows what to fix first.

Job 4: Plan remediation

When an object cannot migrate cleanly, I want clear remediation options so that I can decide whether to rewrite, convert, wait for roadmap support, or use a workaround.

Job 5: Validate cutover

When we complete migration, I want proof that data, pipelines, reports, and security checks match expected behavior so that I can approve production cutover.

6. Current Microsoft tools

Microsoft already provides several migration-related tools and experiences:

Fabric Data Warehouse Migration Assistant

Helps migrate Azure Synapse dedicated SQL pools and other SQL platforms into Fabric Data Warehouse. It supports DACPAC-based migration and a direct source connection path in preview.

Synapse Spark to Fabric Spark Migration Assistant

A public preview tool that migrates Spark pools, notebooks, Spark job definitions, and lake databases. It does not move the data and has limitations around VNet workspaces, non-Delta formats, custom libraries, Spark configurations, and executor settings.

Synapse pipeline migration experience

A preview migration experience for moving Synapse pipelines to Fabric Data Factory. Microsoft documentation notes that activities can remain unmapped or deactivated when required Fabric artifacts or mapped connections are missing.

Azure Key Vault references in Fabric

Fabric supports Azure Key Vault references for connections, but there are limitations. For example, virtual network data gateway connections are not yet supported, Fabric Lineage view is not available for AKV references, and Modern Get Data does not support creating connections authenticated with AKV references.

Fabric Warehouse source control

Fabric Warehouse source control exists in preview, but Microsoft documentation lists limitations. One important limitation is that some ALTER TABLE changes in a database project can drop and recreate a table during deployment, which may cause data loss.

7. What is missing

The missing product layer is not another individual migration tool.

The missing layer is a unified migration confidence experience that combines:

- Estate discovery
- Cross-workload dependency mapping
- Readiness scoring
- Blocker classification
- Remediation planning
- Cutover validation
- Executive reporting

The customer should not have to manually stitch this together from separate migration reports, docs, scripts, spreadsheets, and support conversations.

8. MVP scope

The MVP should focus on assessment and confidence before cutover.

In scope

- Estate overview
- Asset inventory
- Readiness score
- Dependency graph
- Blocker list
- Remediation plan
- Cutover validation checklist
- Executive report export
- Simulated data model for prototype

Out of scope for MVP

- Real Azure API integration
- Real customer telemetry
- Real authentication

- Automated code conversion
- Full migration execution
- Cost simulator based on real billing data
- Real-time monitoring integration
- Production-grade dependency parser

For the prototype, all data is simulated and clearly labeled.

9. Prototype scenario

Customer: Contoso Financial Services

Industry: Banking and financial services

Current platform: Azure Synapse Analytics

Target platform: Microsoft Fabric

Migration stage: Assessment and pre-cutover

Risk level: High

Data type: Simulated enterprise estate metadata

Simulated estate

- 145 Synapse pipelines
- 60 Synapse notebooks
- 140 SQL scripts and stored procedures
- 214 external tables
- 38 linked services and connections
- 72 Power BI semantic models and reports
- 6 Spark pools
- 4 dedicated SQL pools
- 18 serverless SQL databases or views
- 12 private networking dependencies
- 46 Key Vault references or secrets
- 9 CI/CD workflows

Simulated readiness score

Overall Migration Readiness Score: 61/100

Category scores:

- Warehouse compatibility: 68
- Pipeline migration readiness: 54
- Spark and notebook readiness: 72
- Security and private networking readiness: 49
- Monitoring and operations readiness: 52
- BI and report dependency readiness: 76
- Cutover validation readiness: 58

10. Functional requirements

10.1 Overview dashboard

The overview should show:

- Overall readiness score
- Total estate assets
- High-risk blockers
- Remediation item count
- Migration stage
- Top blockers
- Readiness by workload

Customer question answered:

What is the current migration status?

10.2 Estate inventory

The inventory should show all discovered assets across workloads.

Required columns:

- Asset name
- Asset type
- Workload
- Migration status
- Dependencies
- Risk note
- Suggested next step

Supported filters:

- Workload
- Risk level
- Migration status
- Asset type
- Owner

Customer question answered:

What do we have?

10.3 Readiness score

The readiness score should show an overall score and category-level scores.

Score dimensions:

- Asset compatibility

- Dependency completeness
- Security readiness
- Monitoring coverage
- Validation status

Thresholds:

- 0 to 49: Critical
- 50 to 74: Needs work
- 75 to 89: Near ready
- 90 to 100: Ready

Customer question answered:

How ready are we?

10.4 Dependency graph

The dependency graph should show how assets connect across workloads.

Example chain:

External table → Pipeline → Notebook → Power BI report

The graph should highlight:

- High-risk nodes
- Downstream impacted assets
- Blocked dependency chains
- Assets with unmapped connections

Customer question answered:

What depends on what?

10.5 Blocker list

The blocker list should show migration risks grouped by severity.

Blocker fields:

- Blocker title
- Risk level
- Affected workload
- Affected asset count
- Status
- Why it matters
- Suggested remediation
- Owner
- Estimated effort

Customer question answered:

What will stop us?

10.6 Remediation plan

The remediation plan should turn blockers into action items.

Required fields:

- Action
- Priority
- Workload
- Owner
- Status
- Effort estimate
- Dependency
- Expected readiness impact

Customer question answered:

What should we do next?

10.7 Cutover validation

The cutover validation dashboard should prove whether the migration worked.

Validation categories:

- Row count checks
- Schema checks
- Query result comparison
- Pipeline run validation
- Deactivated activity check
- Power BI refresh validation
- Key Vault and connection validation
- Monitoring and alerting validation
- Security and private networking validation

Customer question answered:

Did the migration work?

10.8 Executive report

The executive report should summarize:

- Overall readiness
- Current risk
- Top blockers
- Estimated remediation timeline

- Business impact
- Go or no-go recommendation
- Next actions

Customer question answered:

What do leaders need to know?

11. Blocker taxonomy

High-risk blockers

These block production cutover.

Examples:

- External tables over Parquet/CSV cannot migrate directly
- Pipeline activities are deactivated
- Linked services are unmapped
- Security access path is not validated

Medium-risk blockers

These do not always block cutover, but they create operational risk.

Examples:

- Key Vault references need compatibility check
- CI/CD deployment safety needs review
- Monitoring parity is incomplete
- Custom Spark libraries need manual validation

Low-risk blockers

These are mostly documentation, cleanup, or optimization tasks.

Examples:

- Report owner missing
- Naming mismatch
- Non-critical validation check pending

12. Top blockers in prototype

Blocker 1: External tables over Parquet/CSV

Affected assets: 214

Risk: High

Status: Planned or not fully supported as external table parity

Why it matters: These objects may not migrate directly from Synapse patterns into Fabric Warehouse.
Remediation options:

- Convert to Delta tables
- Use Lakehouse shortcuts
- Use OPENROWSET views where acceptable
- Wait for roadmap support if external-table governance parity is required

Blocker 2: Linked services need Fabric connection mapping

Affected assets: 38

Risk: High

Why it matters: Pipeline activities using unmapped connections can remain deactivated.

Remediation options:

- Map each linked service to a Fabric connection
- Validate authentication
- Confirm Key Vault references
- Reactivate affected activities

Blocker 3: Pipeline activities deactivated after test migration

Affected assets: 23 activities

Risk: High

Why it matters: Pipeline orchestration cannot be trusted until these activities are mapped, activated, and tested.

Remediation options:

- Migrate Spark artifacts first
- Map Fabric notebooks and Spark job definitions
- Reactivate activities
- Run end-to-end validation

Blocker 4: Private networking dependencies need validation

Affected assets: 12 dependencies

Risk: Medium to high

Why it matters: Restricted data customers need confidence that secure access patterns still work.

Remediation options:

- Validate Private Link and firewall paths
- Confirm gateway and VNet behavior
- Test Fabric workload access paths
- Review compliance requirements

Blocker 5: Key Vault references need compatibility check

Affected assets: 46 references

Risk: Medium

Why it matters: AKV exists in Fabric, but not all connection patterns are supported in the same way.
Remediation options:

- Validate supported connectors
- Confirm secret rotation behavior
- Check workspace and tenant setup
- Test dependent connections

Blocker 6: CI/CD workflows need deployment safety review

Affected assets: 9 workflows

Risk: Medium

Why it matters: Enterprise teams need safe schema deployment, rollback, and data-loss checks.

Remediation options:

- Add approval gates
- Add schema diff review
- Add rollback plan
- Add data-loss checks before deployment

13. Readiness scoring logic

The readiness score is calculated from five dimensions:

1. Asset compatibility
2. Dependency completeness
3. Security readiness
4. Monitoring coverage
5. Cutover validation status

Example scoring logic:

- Start with 100 points
- Subtract points for high-risk blockers
- Subtract points for unmapped dependencies
- Subtract points for unvalidated security paths
- Subtract points for failed validation checks
- Add points when remediation actions are completed

Example weight:

- Asset compatibility: 30%
- Dependency completeness: 25%
- Security readiness: 20%
- Monitoring readiness: 10%
- Cutover validation: 15%

The score should be understandable to non-technical stakeholders. The goal is not mathematical perfection. The goal is to create a shared decision-making model.

14. Success metrics

Product success metrics

- Percentage of assets discovered
- Percentage of dependencies mapped
- Number of blockers identified before cutover
- Number of high-risk blockers resolved
- Readiness score improvement over time
- Percentage of validation checks passed
- Time from assessment to cutover-ready state
- Reduction in deactivated pipeline activities
- Reduction in post-cutover incidents

Business success metrics

- Higher Fabric migration conversion rate
- Faster enterprise migration planning
- Lower support burden for Microsoft field teams
- Higher customer trust in Fabric migration
- Lower risk of failed or delayed migrations
- Better executive visibility into migration status

15. Risks and mitigations

Risk 1: Microsoft already has migration tools

Mitigation: Position this product as a confidence layer, not a replacement migration assistant.

Risk 2: Data quality may be incomplete

Mitigation: Show confidence levels for discovered assets and allow manual review.

Risk 3: Some gaps may be fixed by Microsoft roadmap

Mitigation: Classify blockers as supported now, preview, planned, unsupported, or customer remediation.

Risk 4: The prototype uses simulated data

Mitigation: Clearly label all sample data as simulated. Explain that real product data would come from Synapse metadata, Fabric workspace APIs, migration reports, and telemetry.

Risk 5: Product may become too technical

Mitigation: Keep the user experience focused on readiness, blockers, action plans, and cutover decisions.

16. Phased roadmap

Phase 1: Assessment MVP

- Estate inventory
- Readiness score
- Blocker list
- Remediation plan
- Executive report

Phase 2: Dependency intelligence

- Cross-workload dependency graph
- Downstream impact analysis
- Owner and team mapping
- Risk-based prioritization

Phase 3: Validation workflow

- Row count checks
- Query comparison
- Pipeline run validation
- Power BI refresh validation
- Security and connection checks

Phase 4: Microsoft field integration

- Customer success view
- Portfolio-level migration health
- Customer blocker trends
- Playbook recommendations

Phase 5: Advanced intelligence

- Cost and performance risk estimation
- Automated remediation suggestions
- Roadmap-aware blocker tracking
- Migration wave planning

17. Non-goals

This product should not try to:

- Replace Fabric Migration Assistant
- Replace Spark Migration Assistant
- Replace Synapse pipeline migration
- Automatically rewrite every workload
- Claim access to private Microsoft telemetry
- Claim official Synapse customer counts
- Claim a public global Synapse data volume
- Say Synapse is deprecated
- Say Fabric cannot support enterprise workloads

18. Interview narrative

60-second explanation

While preparing for the Microsoft Azure Data PM role, I researched what actually blocks enterprise customers from moving from Synapse to Fabric. I found that Microsoft already has several migration tools, so I did not want to build another generic migration assistant. The deeper problem is migration confidence. Customers need to know what they have, what depends on what, what will break, what needs remediation, and how to prove the migration worked after cutover. So I built a prototype called Fabric Migration Confidence Center. It turns a Synapse estate into an inventory, dependency graph, readiness score, blocker list, remediation plan, and validation dashboard. The prototype uses simulated data, but the workflow is grounded in public Microsoft docs and community pain points.

If asked: Does Microsoft already have migration tools?

Yes, and that is exactly why I framed the product carefully. Microsoft has workload-specific migration tools. This concept does not replace them. It sits above them as a readiness and validation layer across the full estate.

If asked: Is this real data?

No. The prototype uses simulated enterprise estate data. I labeled it clearly. The purpose is to demonstrate the product workflow. In a real Microsoft product, the data could come from Synapse metadata, Fabric workspace APIs, migration reports, Azure Monitor logs, and customer configuration scans.

If asked: Why would Microsoft care?

Because migration confidence affects Fabric adoption. If customers are unsure what will break, they delay migration. A confidence layer can reduce support burden, speed up migration planning, improve customer trust, and help Microsoft field teams prioritize the right blockers.

If asked: What would you validate next?

I would interview Azure/Fabric solution architects, Microsoft field engineers, and enterprise data platform leads. I would ask where migrations get stuck, which blockers show up most often, and which readiness signals they need before approving cutover.

19. Source links

Microsoft Fabric Data Warehouse Migration Assistant:

<https://learn.microsoft.com/en-us/fabric/data-warehouse/migrate-with-migration-assistant>

Migration with direct connection preview:

<https://learn.microsoft.com/en-us/fabric/data-warehouse/migrate-using-connection-to-source-system>

Synapse Spark to Fabric Spark Migration Assistant:

<https://learn.microsoft.com/en-us/fabric/data-engineering/synapse-to-fabric-spark-migration-assistant>

Synapse pipeline migration to Fabric Data Factory:

<https://learn.microsoft.com/en-us/azure/data-factory/how-to-upgrade-your-azure-synapse-analytics-pipelines-to-fabric-data-factory>

Fabric Data Factory limitations:

<https://learn.microsoft.com/en-us/fabric/data-factory/data-factory-limitations>

Azure Key Vault references in Fabric:

<https://learn.microsoft.com/en-us/fabric/data-factory/azure-key-vault-reference-overview>

Fabric Warehouse source control limitations:

<https://learn.microsoft.com/en-us/fabric/data-warehouse/source-control>

Fabric external tables over Parquet/CSV idea:

<https://community.fabric.microsoft.com/t5/Fabric-Ideas/Support-external-tables-for-parquet-csv-in-Fabric-DW/idc-p/4912879>

Reddit example on Synapse external table migration:

<https://www.reddit.com/r/MicrosoftFabric/comments/1oronno/>

Reddit example on Synapse to Fabric pipeline migration:

<https://www.reddit.com/r/dataengineering/comments/1nffjfc/>

Reddit example on Synapse notebook migration limit discussion:

https://www.reddit.com/r/MicrosoftFabric/comments/1t3zecn/how_to_migrate_notebooks_from_synapse/